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REVIEW PAPER

Low-Level Light Therapy in Postherpetic Neuralgia: A Review of Mechanistic Insights and Clinical Implications

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Abstract

Known as the most prevailing complication of herpes zoster (HZ), postherpetic neuralgia (PHN) can lead to long-term skin discomfort. Current medications for PHN remain suboptimal with incomplete relief and potential side effects. Low-level light therapy (LLLT) has been increasingly explored as a non-invasive modality for treating PHN, but the mechanistic basis and clinical implications have not been synthesized in a translational framework. In this review, we explore the pathophysiological mechanisms of PHN and the biological mechanisms of LLLT, integrating current mechanistic insights with available clinical evidence. We summarize the experimental and clinical literature evaluating LLLT for PHN and related neuropathic pain models, mainly focusing on mechanisms across peripheral tissues, dorsal root ganglia, and neuro-immune aspects, as we evaluate therapeutic parameters and outcomes from available clinical studies. Current evidence suggests that PHN is associated with dorsal root ganglion (DRG) inflammation, peripheral cutaneous neuroinflammation, loss of cutaneous innervation, and ion channel dysregulation. These changes comprehensively lead to hyperexcitable neurons with a lower sensory threshold. Across neuropathic pain models, LLLT elevates mechanical and thermal pain thresholds, attenuates inflammatory cytokine signaling, and promotes an immune cell phenotype shift toward an anti-inflammatory profile including M1-to-M2 macrophage polarization. LLLT also stabilizes membrane excitability by restoring Na^+/K^+-ATPase activity and modulating calcium dynamics, while supporting peripheral nerve repair via axonal regeneration. Clinically, administering LLLT to patients with PHN resulted in significant pain reduction, and early intervention may also reduce its incidence. By integrating experimental and clinical research, LLLT serves as a promising approach for PHN owing to its efficacy for anti-inflammation, neuronal regeneration and ion modulation.

Graphical Abstract

Low-level light therapy for postherpetic neuralgia. Summary of the proposed mechanisms of LLLT in PHN. PHN involves dorsal root ganglion inflammation, ion channel dysfunction, and loss of cutaneous nerve fibers. LLLT may alleviate neuropathic pain by reducing neuroinflammation, modulating ion channels, activating Schwann cells, and promoting axonal regeneration

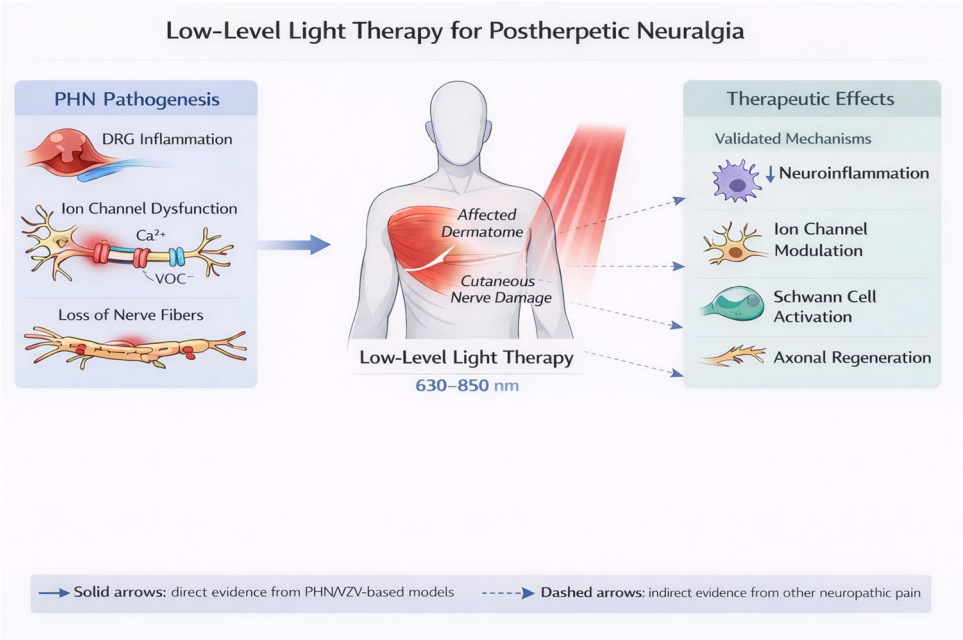
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Keywords Postherpetic neuralgia · Low-level light therapy · Photobiomodulation · Neuropathic pain · Neuroinflammation · Peripheral nerve regeneration

Abbreviations

LLLT	Low-level light therapy
PHN	Postherpetic neuralgia
HZ	Herpes zoster
VZV	Varicella-zoster virus
RCT	Randomized controlled trials
AlGaAs	Aluminum gallium-arsenide
TRPV1	Transient receptor potential vanilloid-1
IL	Interleukin
DRG	Dorsal root ganglion
nAChRs	Nicotinic acetylcholine receptors
HIF-1α	Hypoxia-inducible factor 1α
TNF-α	Tumor-necrosis factor
bFGF	Basic fibroblast growth factor
NGF	Nerve growth factor
VAS	Visual analogue scale
NPRS	Numerical pain rating scale
TENS	Transcutaneous electrical nerve stimulation

Introduction

Herpes zoster(HZ), also known as shingles, results from the reactivation of varicella-zoster virus (VZV) following a decline in VZV-specific immunity. Patients affected by this neurocutaneous disease often experience symptoms such as acute pain, abnormal sensations, and the incidence of maculopapular lesions (Saguil et al. 2017). In the general population, the incidence rate of HZ throughout the lifetime is around 20–30%, with the number increasing up to 50% in people over 85 years of age. Cases of HZ in adults ≥ 65 years old rose from 2.83 million in 1992 to 6.52 million in 2021, which is a major burden for healthcare systems around the globe (Li et al. 2025; Lin et al. 2024). Known as the most common long-term complication of HZ, postherpetic neuralgia (PHN) is defined by persistent neuropathic pain in a dermatomal distribution following an acute onset of HZ episode, lasting beyond the healing of the primary rash. In this review, we adopt a time-based definition of PHN as pain persisting for ≥90 days after rash onset (or zoster diagnosis), consistent with commonly used clinical criteria (Kim et al. 2022; Drolet et al. 2010). PHN occurs more frequently in immunocompromised individuals, and its incidence rises sharply with advancing age, reflecting the pivotal role of declining cell-mediated immunity in viral reactivation and subsequent nerve injury. This neuro-debilitating complication can be further recognized by sensory alterations such as numbness and dysesthesias, easily evoked pain such as allodynia and hyperalgesia, or even pruritic lesions noted at the previous dermal region of the HZ outburst (Gruver and Guthmiller 2025; Park et al. 2022).

Current treatment modalities for PHN include tricyclic antidepressants, gabapentinoids, opioids, and topical agents, but those often provide only temporary relief and are limited by adverse effects. Despite multimodal therapy, their

efficacy remains suboptimal, highlighting the urgent need to identify new therapeutic approaches (Huerta et al. 2023). Low-level light therapy(LLLT), also known as photobiomodulation and low-level laser therapy, is a non-invasive technique that utilizes low-intensity red or near-infrared light to modulate biological processes. With reported potential benefits of dampening neuropathic pain, LLLT has emerged as a novel therapeutic option worthy of clinical evaluation (de Andrade et al. 2016).

Multiple clinical studies have suggested the effect of LLLT to alleviate the severe symptoms that patients with PHN experience. For instance, in 2 randomized controlled trials (RCTs), LLLT using an aluminum gallium-arsenide (AlGaAs) diode laser, whether used alone or in combination with other medications, has shown significant improvements in pain reduction (Anwar et al. 2016; Mahmed ML 2023). In addition, in a retrospective observational study, LLLT was shown to be effective in alleviating pain both with acute (within 5 days of rash onset) and with subacute (within 6–14 days of rash onset) administration. With acute administration of LLLT, the incidence of PHN could even be reduced to 0% (Chen et al. 2016). Although LLLT has not yet been adopted as a standard treatment for PHN, clinical applications have already revealed significant outcomes. A systematic review and meta-analysis indicates that LLLT may outperform conventional treatments in reducing PHN-related pain, suggesting that it may represent a promising adjunctive therapeutic approach (Sathya Siva et al. 2024).

Despite the fact that numerous clinical studies conducted revealed satisfactory outcomes, the underlying mechanism of LLLT in treating PHN has remained unspecified. This review discusses the neurobiological processes modulated by LLLT in PHN, thereby providing mechanistic insights into PHN treatment. Importantly, the currently available mechanistic evidence is heterogeneous and varies in its direct relevance to PHN. To avoid overinterpretation, the mechanistic framework discussed in this review is organized into three categories of evidence. Level 1 refers to direct mechanistic evidence derived from PHN patients or VZV-based experimental models. Level 2 refers to indirect mechanistic evidence derived from other neuropathic pain models that may be relevant to PHN pathobiology. Level 3 refers to mechanistic hypotheses that remain speculative and have not yet been directly validated in PHN or VZV-based systems. Throughout this review, each mechanistic subsection is labeled according to this hierarchy to distinguish established observations from indirect inferences and emerging hypotheses.

Methods (Literature Search Strategy)

This narrative review was conducted through a structured literature search to identify relevant studies on LLLT in PHN and related mechanistic pathways. Electronic databases including PubMed, Embase, and Google Scholar were searched for articles published up to 2026. Keywords used in the search included combinations of “postherpetic neuralgia,” “herpes zoster,” “varicella-zoster virus,” “low-level light therapy,” “photobiomodulation,” “laser therapy,” “neuropathic pain,” “inflammation,” and “nerve regeneration.”

Both clinical and experimental studies were considered, including randomized controlled trials, observational studies, and relevant in vivo and in vitro mechanistic research. Articles were selected based on their relevance to the clinical effects or biological mechanisms of LLLT in PHN or related neuropathic pain conditions. Reference lists of selected articles were also reviewed to identify additional relevant studies.

Definition and Pathophysiology of PHN

Definition

The diagnosis of PHN is universally based on the evaluation of history taking and physical examination. For history taking, a previous onset of HZ establishes a prerequisite condition for diagnosing PHN. This dermatomal neuralgia can be further categorized and characterized as persistent burning sensations, pruritus, paresthesia, dysesthesia, allodynia, hyperalgesia, and lancinating sensations. Clinically, PHN is typically defined using a time-based threshold after rash onset; in this review we use $\geq$90 days as the primary definition (Kim et al. 2022; Drolet et al. 2010). Upon examination, the affected dermatome may show residual cutaneous changes, such as scarring or altered pigmentation. In some cases, autonomic changes (e.g., localized sweating or temperature variation) may also occur (Mallick-Searle et al. 2016; Gruver and Guthmiller 2025). The underlying mechanism of PHN has not been fully understood. Current studies indicate that its pathophysiology is multifactorial, broadly involving peripheral nerves, sensory ganglia and cutaneous regions (Truini et al. 2013; Kinchington and Goins 2011).

Damage to Sensory Ganglia

(Level 1: Direct Mechanistic Evidence from VZV-based Models)

After infecting the host, VZV will remain latent in places such as cranial nerve ganglia, dorsal root ganglia (DRG),

enteric and autonomic ganglia, and can then reactivate in immunocompromised states, leading to HZ. During viral reactivation, the virus replicates in the sensory ganglia. Viral replication and intra-ganglionic spread, often involving adjacent neurons, allow anterograde transmission to peripheral nerves. The process of viral spread then further triggers ganglionic inflammation, neuronal apoptosis, and axonal degeneration (Gershon et al. 2015; Kinchington and Goins 2011). This is accompanied by a robust immune response characterized by lymphocytic infiltration, the release of proinflammatory cytokines including interleukin (IL) −1β, IL-6 and tumor necrosis factor (TNF) -α and the sustained inflammation, exacerbating neuronal injury by causing the hyperexcitability of DRG neurons (Jang and Garraway 2024). The resulting disruption of afferent sensory pathways, including the degeneration of small fiber afferents and damage to central sensory roots, leads to abnormal pain signaling. Over time, central sensitization and the progressive loss of inhibitory interneurons lead to maladaptive plasticity that perpetuates neuronal hyperexcitability. These changes establish a chronic pain state marked by allodynia and hyperalgesia, even in the absence of active viral replication (Zerboni and Arvin 2015; Kinchington and Goins 2011).

Damage to Cutaneous Nerves and Inflammatory Responses of Dermal Regions

(Level 1: Direct Mechanistic Evidence from VZV-based Models)

Other than debilitating sensory ganglia, VZV travels and spreads further to the periphery, often involving cutaneous fibers innervating skin dermatomes. Affected nerve fibers reveal changes such as degenerations, impaired functions or even the loss of innervation (Kinchington and Goins 2011). Studies have demonstrated that greater loss of cutaneous innervation is correlated with abnormal nociceptor sensitization and increased pain severity in PHN (Oaklander 2001; Rowbotham et al. 1996; Lauria et al. 2009). Furthermore, patients with PHN also demonstrated a marked loss of myelinated axons (Watson et al. 1991).

Keratinocytes play an important role in the epidermal barrier. VZV replicates and spreads through the epidermis in association with inhibiting the apoptosis of keratinocytes, as histological evidence showed its capacity to overcome antiviral responses and generate vesicles in epidermal regions (Ku et al. 2004; Tommasi and Breuer 2022; Gerada et al. 2018). This epidermal invasion could be facilitated by the reactivated VZV's predominant infection of CD4+T cells, thereby promoting the release of cytokines and the recruitment of innate and adaptive immune cells into the infected dermal region (Zhang and An 2007; Wang et al. 2005). This reveals that interactions between immune cells and inflammatory responses around dermal regions greatly contribute to PHN (Ku et al. 2004; Torigo et al. 2000; Liu et al. 2024).

Ion Channel Dysregulation

(Level 1: Direct Mechanistic Evidence from VZV-based Models)

The dysregulation of ion channels observed in PHN often leads to the hyperexcitability state of affected neurons, causing central and peripheral sensitization (Kinchington and Goins 2011; Ma et al. 2024). Sodium ion channels, in particular, have been proven to constitute neuropathic pain in a VZV-infected rat model (Garry et al. 2005). A study by Kennedy et al. further illustrated that PHN-associated VZV isolates enhance neuronal excitability by upregulating TTX-sensitive sodium currents through Nav1.6 and Nav1.7 channels, potentially contributing to the pathogenesis of PHN (Kennedy et al. 2013). Evidence (Kennedy et al. 2013) has shown an upregulation of Nav1.8 sodium channel expression in patients infected with VZV, which is associated with a lowered threshold for mechanical stimulation and the subsequent development of neuropathic pain. On the other hand, other studies have revealed that PHN is associated with the upregulated expression of the α2δ1 calcium channel in neurons(Garry et al. 2005). By systematically administering gabapentin, which inhibits the release of synaptic excitatory neurotransmitters, the hyperexcited sensitization reflex was attenuated in patients with PHN (Akopian et al. 1996; Garry et al. 2005). Transient receptor potential vanilloid-1 (TRPV1) has also been linked to acute pain in PHN, and capsaicin patches have been utilized to alleviate neuropathic pain through defunctionalization of the TRPV1 channel (Laklouk and Baranidharan 2016; Han et al. 2016). These findings collectively indicate that ion channel dysregulation plays a pivotal role in the development and maintenance of PHN by lowering pain thresholds.

Mechanisms of LLLT in Treating PHN

PHN is widely recognized as a classical form of chronic neuropathic pain, sharing many pathophysiological features with other peripheral neuropathic pain disorders. In PHN patients, persistent pain can be triggered by prolonged peripheral neuropathic inflammation (Hadley et al. 2016; Liu et al. 2024). LLLT, which has a profound potential to promote tissue regeneration, modulate inflammatory response, and attenuate neuropathic pain (Yeh et al. 2019; da Silva et al. 2023), offers a promising non-pharmacological approach

(Fig. 1). Therefore, since PHN shares numerous features observed in neuropathic pain, LLLT may exert potential therapeutic effects on PHN by targeting similar inflammatory and neuroplastic pathways.

Establishing a VZV-induced Pain Model

(Level 1: Direct Mechanistic Evidence from VZV-based Models)

Experimental studies have established reproducible VZV-induced pain models that provide direct mechanistic insight into PHN-relevant pathophysiology. First, the subcutaneous injection of cell-associated VZV was performed into the hind paws of rats, after which pain behaviors and mechanical stimuli tests were performed to objectively assess the extent of somatosensory change. By conducting paw withdrawal tests, a significantly lower threshold was noted at the ipsilateral paw of the VZV inoculation, which induced a mechanical allodynia and thermal hyperalgesia within 3–5 days, lasting up to 14 weeks. DRG, deemed a crucial site for viral replication and reactivation in humans, was detected with VZV genomes and viral transcripts in this model. Peripherally, the loss of epidermal neuronal axons innervation was also identified. (Ou et al. 2023) concluded that this VZV-induced hypersensitivity closely resembles clinical PHN in 4 aspects: changes in nerve systems, inflammatory responses, pain-related genes and the upregulation of ion channels (α2δ1 calcium channel, Nav1.3 and Na1.8). Although rats are not the natural host of VZV, the clinical course and molecular changes in their nervous system after inoculation of the virus still mirror human disease, showing virus transport to DRG, transient low-copy viral genomes, and prolonged mechanical allodynia consistent with DRG-centric mechanisms seen in PHN. These models provide important mechanistic insights into PHN pathophysiology (Dalziel et al. 2004; Ou et al. 2023; Kinchington and Goins 2011; Guedon et al. 2015).

LLLT Modulates Allodynia and Hyperalgesia by Elevating Sensory Threshold in Experimental Neuropathic Pain Models

(Level 2: Indirect Mechanistic Evidence from Other Neuropathic Pain Conditions)

The overall hypersensitivity in PHN can be categorized as mechanical allodynia and thermal hyperalgesia. The former could be assessed by the significant lowering of the threshold of hind paw withdrawal, while the latter was characterized by a reduction in paw withdrawal latency in response to noxious heat stimulation (Kinchington and Goins 2011). Although direct evidence in PHN-specific models remains limited, multiple non-VZV neuropathic pain models suggest that LLLT may attenuate both forms of hypersensitivity. In numerous experimental neuropathic pain animal models such as the hind-paw ischemia reperfusion model, the spared nerve injury model and the post-operative pain model, LLLT has been found to alleviate hyperalgesia and allodynia by elevating the sensory threshold (Prianti et al. 2014; Mojarad et al. 2018; Han et al. 2024; Pigatto et al. 2020). Furthermore, in a chronic constriction injury rodent model, LLLT was shown not only to alleviate allodynia but also to dampen its development (Chacur et al. 2024). Collectively, LLLT may help restore sensory balance and normalize dysregulated sensory thresholds by dampening dysregulated nociceptive signaling. However, further in vivo experiments involving direct VZV-based studies are still required to confirm the modulatory effect of LLLT for VZV-infected subjects.

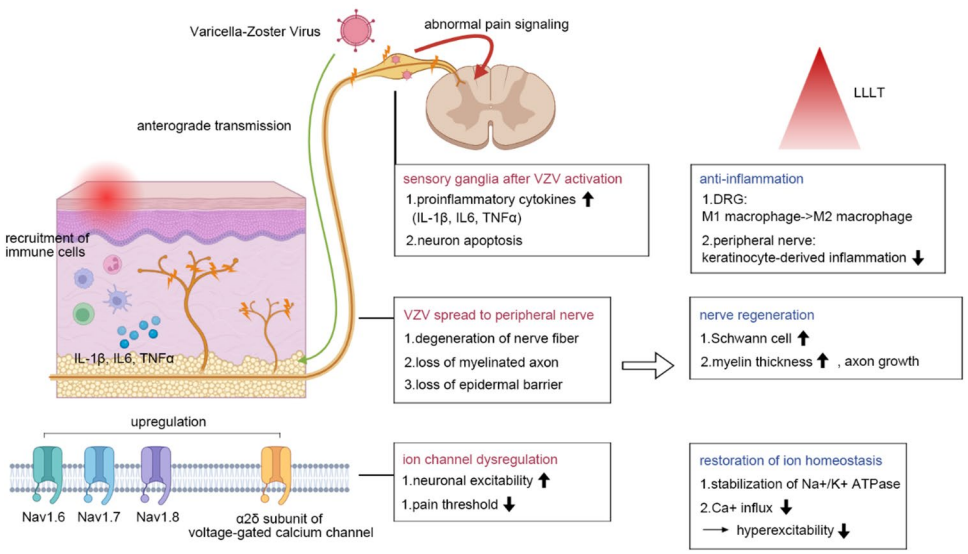


Fig. 1 Overview of PHN pathophysiology and the proposed therapeutic mechanisms of LLLT. The reactivation of VZV causes ganglionic inflammation. With its migration via peripheral nerves to the epidermal barrier, VZV further induces ion channel disruption and nerve fiber degeneration, ultimately promoting neuropathic pain. LLLT ameliorates these changes through anti-inflammatory effects (M1→M2 shift and reduced keratinocyte cytokines), promotion of nerve regeneration (Schwann cell activation, axon and myelin growth), and restoration of ion homeostasis (stabilized Na^+/K^+- ATPase and reduced Ca^{2+} influx), reducing allodynia and hyperalgesia

LLLT Alleviates Neuropathic Pain by Modulating the Inflammatory Response

(Level 2: Indirect Mechanistic Evidence from Other Neuropathic Pain Conditions)

The inflammatory cascade of PHN can be broadly categorized into 2 mechanisms: (1) Ganglionic inflammation within the DRG, and (2) Peripheral inflammation involving cutaneous and perineural tissues (Wang et al. 2005; Kinchington and Goins 2011; Zhang and An 2007). However, the mechanistic evidence supporting LLLT-mediated anti-inflammatory effects differs in strength depending on the biological context.

DRG Level Modulation by LLLT Within the DRG, PHN-associated inflammation is characterized by macrophage activation, lymphocytic infiltration, and proinflammatory cytokine release. In non-PHN experimental neuropathic pain models, LLLT manifested a distinct immunomodulatory effect at the level of the DRG. Specifically, LLLT induced a significant shift from M1 macrophages (pro-inflammatory) toward M2 macrophages (anti-inflammatory). M2 macrophages are associated with characteristics such as the secretion of anti-inflammatory cytokines and the promotion of tissue repair, indicating that LLLT may attenuate neuropathic pain by driving macrophages toward a neuroprotective profile (Kobiela Ketz et al. 2017; Luo et al. 2024). Additional evidence from neuropathic pain models also shows that LLLT significantly suppresses the expression of hypoxia-inducible factor 1α (HIF-1α), TNF-α, IL-1β, and IL-6 at peripheral nerve injury sites (Hsieh et al. 2012).

Peripheral Cutaneous Level Modulation by LLLT Epidermal barrier disruption may contribute to dysregulated immune responses. At the cutaneous level, keratinocytes are characterized as one of the major targets of VZV and are responsible for secreting inflammatory cytokines under immunological stress. In non-PHN experimental settings, LLLT was proven to counteract keratinocyte-driven inflammation by reducing TNF-α, IL-6, and IL-8 expression via activation of the Nrf2 pathway (Salman et al. 2023; Black et al. 2009; Jones et al. 2014).

These observations raise the possibility that LLLT may mitigate cutaneous neuroinflammatory processes relevant to PHN. However, direct evidence linking LLLT to reduced keratinocyte-driven inflammation in PHN or VZV-infected skin remains limited.

Ion Channel Modulation as a Mechanism of LLLT for PHN

(Level 2: Indirect Mechanistic Evidence from Other Neuropathic Pain Conditions)

As illustrated above, the dysregulation of certain ion channels, such as sodium and calcium channels, greatly contributes to the pathophysiology of PHN. LLLT is believed to modulate the activity of ion pumps and channels (e.g., Na^+/K^+-ATPase, sodium, potassium, and TRP channels), suggesting potential analgesic effects (Zhang, Zhang et al. 2024a, b). For calcium balance, LLLT modulates calcium homeostasis through both ion channel regulation and intracellular signaling pathways. In a bupivacaine-induced nerve injury model, AlGaAs laser therapy with a wavelength of 904 nm restored neuromuscular junction morphology and increased expression of the ε-subunit of nicotinic acetylcholine receptors (nAChRs), which stabilizes interactions between nerve and muscle via the modulation of calcium influx (Castro et al. 2022; Pissulin et al. 2017). In a diabetic neuropathy model, LLLT at the same wavelength (904 nm) reduced hyperalgesia by modulating calcium dynamics and attenuating the excessive Ca^{2+} influx probably mediated through TRPV1-expressing nociceptive neurons, thereby reducing calcium-dependent neuronal hyperexcitability and pain signaling (Vieira et al. 2022). Additional experimental cellular studies have also found a correlation between LLLT and the stabilization of membrane potential. A 660 nm LED array was found to restore Na^+/K^+-ATPase function, counteracting the pump inhibition that normally leads to membrane depolarization and ionic imbalance (Rhee et al. 2019). Currently, there is no evidence indicating that LLLT directly suppresses the expression of specific ion channels. Taken together, these findings support the hypothesis that LLLT may reduce neuronal hyperexcitability by stabilizing membrane potential and modulating ionic homeostasis. Nevertheless, direct evidence for this mechanism in PHN or VZV-based models is currently lacking.

LLLT in Repairing Peripheral Nerve and Prompting Schwann Cell Regeneration

(Level 2: Indirect Mechanistic Evidence from Other Neuropathic Pain Conditions)

LLLT has been repeatedly shown to promote peripheral nerve regeneration by stimulating the behavior of Schwann cells, thickening axon diameters, enhancing axon quantity and releasing neuro-regenerative substances (Rosso et al. 2018; Yazdani et al. 2012; Muniz et al. 2021). In rodent nerve crush models, research studies revealed that LLLT

enhances both structural and functional recovery. This could be the result of accelerated axonal regrowth, increased myelin thickness and elevated levels of growth-associated protein 43 (GAP43)(Wang et al. 2014; Gigo-Benato et al. 2010; Alcântara et al. 2013). On the other hand, in in vitro cellular studies, LLLT (808 nm) stimulates human Schwann cell proliferation and nerve growth factor (NGF) expression (Yazdani et al. 2012), while irradiation with a He-Ne laser (632.8 nm) promotes the expression of basic fibroblast growth factor (bFGF) and NGF in human keratinocytes (Yu et al. 2003). Evidence from facial nerve injury models further showed that LLLT reduces Schwann cell apoptosis and promotes their proliferation via upregulation of the PI3K/Akt signaling pathway, leading to significant axonal regeneration and functional restoration (Li and Wang 2022).

Potential Implications of LLLT-responsive PI3K/Akt Signaling for VZV Latency and Reactivation

(Level 3: Speculative Mechanistic Hypothesis)

Beyond structural restoration, the elevated NGF level induced by LLLT may also exert downstream neuroprotective effects and has been hypothesized to influence signaling pathways implicated in VZV latency and reactivation; however, direct antiviral effects of LLLT have not been demonstrated.By binding to TrkA receptors, NGF activates the PI3K/Akt signaling cascade (Yan et al. 2020), which was previously reported by to suppress VZV reactivation in neuronal cells (Markus et al. 2015). Moreover, in an in vitro study of vitiligo, exposure of NCCmelb4 melanoblasts to a He-Ne laser has been shown to elevate focal adhesion kinase (FAK/pp125) activity, a known upstream regulator of Akt, suggesting an additional entry point for LLLT-mediated PI3K/Akt activation(Yu et al. 2003).

Through convergent stimulation of the FAK- and NGF-mediated PI3K/Akt signaling pathway, LLLT may support neuroprotective and regenerative processes. However, no direct evidence currently demonstrates that LLLT alters VZV latency, replication, or reactivation in PHN patients or VZV-based experimental models. Given the complexity of host–virus interactions governing viral latency, this proposed pathway should be regarded as a hypothesis requiring experimental validation, rather than an established mechanism contributing to the therapeutic effects of LLLT in PHN(Lan et al. 2006; Ye et al. 2024; Markus et al. 2015).

Discussion

LLLT appears to exert its therapeutic effect multi-dimensionally to modulate PHN-related neurobiological alterations. For neuro-inflammatory responses, the inflammation cascade of PHN involves peripheral and ganglionic mechanisms. LLLT demonstrates its immunomodulatory potential by altering the polarization of macrophages from the M1 to M2 phenotype and by downregulating levels of pro-inflammatory cytokines (Zhang et al. 2024a, b). For the dysregulation of ion channel and electrolyte imbalance, LLLT restores the activity of Na^+/K^+-ATPase, regulates the excessive calcium influx and stabilizes the membrane potential. For the profile of neuro-structural damage, LLLT has been reported to stimulate Schwann cell activity, thereby enhancing axonal regeneration. In addition, LLLT has been hypothesized to influence signaling pathways (e.g., FAK/PI3K/Akt) that have been implicated in the regulation of VZV latency; however, direct antiviral effects of LLLT in PHN-relevant contexts have not been established. These alterations brought by irradiation with light therapy may contribute to the improvement of PHN-associated neuropathic damage. Taken together, the multimodal actions of LLLT provide a rationale for further clinical investigation to treat PHN, not only as a symptomatic treatment but also as a potential adjunctive therapeutic approach that addresses the underlying pathophysiology.

This review considered 7 clinical studies associated with the administration of LLLT in PHN, as we sought to compile a thorough coverage of clinical evidence **(**Table 1**)**. Across the included studies, wavelength, treatment frequency, and total number of sessions were consistently reported parameters, representing the most reproducible elements across trials. In a RCT of sciatic-nerve PHN, treatment with 830 nm AlGaAs LLLT once weekly for 3 months reduced the mean visual analogue scale (VAS) from 9.0 ± 0.86 to 2.65 ± 0.59 and the numerical pain rating scale (NPRS)(ranging from 0 to 100 in this study) from 72.5 ± 8.66 to 25.0 ± 4.87, whereas the placebo-laser treated group showed virtually no change (VAS 9.05 → 8.95; NPRS 72.25 → 72.0)(Anwar et al. 2016). In another RCT comparing LLLT with transcutaneous electrical nerve stimulation (TENS), adding 850 nm LLLT to standard medical treatment three times weekly for 4 weeks resulted in an 88.97% reduction in VAS (5.44 → 0.60) and a remarkable increase in pain-pressure threshold (0.45 → 2.80 kg), exceeding the improvements seen with TENS alone. Similarly, a larger double-blind crossover trial of 63 PHN patients that were treated with 830 nm AlGaAs LLLT 2–6 times per week found improvements in VAS ratings (Osamu Kemmotsu et al. 1991). In another double-blind crossover trial of 20 patients with long-standing PHN that were treated with 830 nm AlGaAs LLLT twice weekly,

Table 1 Summary of clinical studies involving LLLT in PHN

Authors	Year	Country	Study design	Population	Laser (Type/Wavelength)	Laser power and mode	Treatment protocol	Control/Co-therapy	Scales	Follow-up timepoints	Main effects
Anwar et al. (2016)	2016	Egypt	RCT (placebo-controlled)	PHN of sciatic nerve(45–65 years old); $N=40$	AlGaAs diode; 830 nm	NR	4 sites along sciatic course; 5 min per site (total 20 min); once weekly × 3 months	Placebo laser	VAS; NPRS	Baseline and 3 months	Significant improvement in sciatic PHN
MAhmed et al. (2023)	2023	Egypt	RCT (active comparator: LLLT vs. TENS)	Patients with PHN(40–80 years old); $N=50$ (LLLT $n=25$; TENS $n=25$)	AlGaAs cluster diode, 850 nm	Dose 3.6 J/cm² per tender point, 1 min/point	20 min/session; 3 times/week × 4 weeks	High-frequency TENS; both groups received standard medical treatment	VAS; PPT (kg) via electronic algometer	Baseline and post-intervention (4 weeks)	Significantly reduced pain and improved pressure pain threshold in PHN patients compared to TENS
Wirya et al. (2023)	2023	Indonesia (Bali)	Retrospective cohort (pre and post LLLT within subjects)	Patients with PHN for 1–12 months(mean age 59.8 ± 15.2); $N=34$	AlGaAs diode; 660 nm (HANIL HLA-2000)	15 mW; scanning 10 × 10 cm area; 1 cm aperture; 15 min/area	16 sessions over 8 weeks (2×/week)	Self-controlled; all received co-therapy (gabapentin + vitamins B1,6,12)	NPRS	Baseline, 4 weeks, 8 weeks (on-treatment)	LLLT significantly reduces pain in PHN patients without adverse effects
Chen et al. (2016)	2016	Taiwan	Retrospective cohort (acute LLLT vs. subacute LLLT vs. no LLLT control)	Patients with HZ; Acute LLLT $n=48$ (≤5 days), Subacute LLLT $n=48$ (days 6–14), Control $n=154$	He–Ne laser; 632.8 nm	50 mW; 2 J/cm²; applied to affected dermatome	6 min/session; every other day(3×/week); patients received ≥ 1 complete session	No LLLT (control); concomitant famciclovir use balanced and adjusted in model	Incidence of PHN	1, 3, 6 months	Early application of LLLT in herpes zoster patients significantly lowers the incidence of PHN

Table 1 (continued)

Authors	Year	Country	Study design	Population	Laser (Type/Wavelength)	Laser power and mode	Treatment protocol	Control/Co-therapy	Scales	Follow-up timepoints	Main effects
Mukhtar et al. (2020)	2020	Pakistan	Prospective uncontrolled case series (pilot)	Patients with PHN (42–82 years old) refractory to meds; $N=15$	AlGaAs diode; 650 nm	Dose 3.6 J/cm² per tender point; 1 min/point	16 sessions over 8 weeks (2×/week)	Self-controlled	VAS pre/post LLLT; qualitative reports of medication reduction and QoL	Baseline and post-16 sessions	No recurrence of pain or abnormality after LLLT
Kemmotsu et al. (1991)	1991	Japan	Double blind crossover trial	Patients with PHN (33–91 years old); $N=63$	AlGaAs cluster diode; 830 nm	Dose 1.2–3 J/cm² per affect area	10–20 min/session; 2-3X/week for outpatients; 4-6X/week for inpatients	Self-controlled	Linear analogue scale	Baseline, immediate and long term	LLLT provides both immediate and long term analgesic effects
Kevin et al. (2004)	2004	UK	Double blind crossover trial	Patients with PHN(55–82 years old); $N=20$	AlGaAs diode laser; 830 nm	Dose 3 J/cm² per tender point; 15 s/point	20 min/session; 2 times/week × 8 weeks	Placebo laser	Linear analogue scale	Baseline and every week	LLLT reduces pain severity and distribution

LLLT: low-level laser therapy; PHN: postherpetic neuralgia; NR: not reported; RCT: randomized controlled study; VAS: visual analogue scale; NPRS: numerical pain rating scale; PPT: pain pressure threshold; TENS: transcutaneous electrical nerve stimulation; QoL: quality of life

all patients achieved reductions in pain intensity and painful area, 95% of them were able to stop their analgesic medications, and 80% maintained those improvements over a 3–6 month follow-up (Moore et al. 2004). Beyond symptomatic relief, a Taiwanese retrospective cohort showed that initiating LLLT within 5 days of rash onset reduced PHN incidence when compared with patients not receiving LLLT (27.3% vs. 8.3%) at 1 month, (14.3% to 2.1%) at 3 months, and (0% vs. 9.7%) at 6 months (Chen et al. 2016). For established, drug-refractory PHN, a Pakistani pilot study using 650 nm LLLT twice weekly for 8 weeks reported that 11 of 15 patients became pain-free (final VAS 0), three improved to mild pain (VAS 2–3), and one improved to moderate pain (VAS 4), with no reported recurrence (Mukhtar et al. 2020). Consistent findings were observed in a 34-patient Indonesian retrospective cohort, where 660-nm AlGaAs LLLT twice weekly for eight weeks reduced the NPRS mean from 8.26 ± 1.31 to 1.74 ± 1.33, allowing gabapentin to be tapered or discontinued in many cases (Wirya AY 2023).

These above-mentioned studies have suggested the efficacy of the clinical application of LLLT in patients with PHN. These clinical observations may be partly explained by the convergent characteristics of an anti-inflammatory profile, analgesics effects and neuronal repair, as these biological mechanisms may be relevant to the clinical improvements observed in these studies. The rapid reduction in pain scores observed in those patients may be consistent with the capacity of LLLT to suppress neuroinflammation through M1→M2 macrophage transition, the attenuation of cytokine-driven sensitization, and the restoration of ion-channel homeostasis. The observed improvement in chronic, long-standing PHN, which is often refractory to pharmacologic therapy, suggests that LLLT may contribute to neurostructural repair processes, including remyelination and axonal regeneration. This specific trait might also contribute to the reduction of the affected dermatome observed in clinical studies (Moore et al. 2004). Moreover, the observed reduction in PHN incidence with early LLLT administration may reflect attenuation of acute neuroinflammation and improved neural repair. A potential antiviral contribution remains speculative (Chen et al. 2016). By integrating the trajectory of biological mechanisms and clinical applications, we aim to integrate available evidence and provide an integrative overview of the potential relationship between PHN pathophysiology and the proposed effects of LLLT. While these clinical findings are encouraging, the relationship between observed clinical outcomes and proposed biological mechanisms should be interpreted with caution. The reduction in pain scores may be consistent with anti-inflammatory and neuromodulatory effects of LLLT described in experimental models; however, most clinical studies do not include mechanistic endpoints, and direct evidence linking these biological processes to clinical outcomes in PHN remains limited. Similarly, improvements observed in chronic or refractory cases may reflect enhanced neural repair, but this association has not been directly demonstrated in human PHN studies. The potential role of LLLT in reducing PHN incidence when administered early in the disease course may be related to modulation of acute neuroinflammation or nerve injury, although causal relationships have not been established.

Future studies are warranted to further clarify the mechanisms and optimize the clinical application of LLLT in PHN. Experimental models that more closely recapitulate VZV-induced neuropathology are needed to directly evaluate its effects on neuroinflammation and neuronal injury. In addition, clinical studies incorporating mechanistic endpoints and standardized irradiation parameters would help link clinical outcomes with underlying biological processes. Longitudinal studies are also required to assess the durability of therapeutic effects. Finally, the hypothesized interaction between LLLT and viral latency pathways, including PI3K/Akt signaling, warrants direct investigation in PHN-relevant models.

LLLT has shown efficacy to reduce pain, lower the incidence of PHN when administered during acute or subacute stages, and achieve meaningful relief even in patients refractory to conventional pharmacological treatments. In addition, current studies have manifested the safety profile and adjunctive efficacy of LLLT, thereby representing a promising therapeutic option for elderly patients or those who cannot tolerate the adverse effects of conventional pharmacologic interventions.

Limitations of Current Clinical Evidence

Although the available clinical studies suggest that LLLT may provide analgesic benefits in patients with PHN, several methodological limitations should be considered. First, the number of randomized controlled trials remains limited, and many studies involve relatively small sample sizes, which may reduce statistical power and generalizability. Second, substantial heterogeneity exists in treatment protocols, including differences in wavelength, power output, fluence, irradiation duration, and treatment frequency, making direct comparison across studies difficult and preventing the identification of optimal therapeutic parameters. Third, some studies are retrospective or lack adequate randomization and blinding, which may introduce potential bias. In addition, outcome measures vary across studies, including differences in pain assessment tools and follow-up duration. Importantly, most clinical studies do not incorporate mechanistic endpoints, limiting the ability to directly link observed clinical improvements with the underlying biological effects

of LLLT. Taken together, these limitations highlight the need for larger, well-designed randomized controlled trials with standardized protocols and integrated mechanistic evaluation.

Conclusion

The sum of the evidence about LLLT reviewed here suggests potential efficacy in modulating the neuroimmune-driven inflammatory response. Combined with its neuroregenerative profile, LLLT may represent a promising non-invasive adjunctive modality for treating medication-refractory PHN. These findings highlight the importance of understanding the underlying mechanism of how LLLT has its therapeutic effects. Our review offers an insightful perspective toward the pathophysiology of PHN and how these neurobiological alterations may be modulated by LLLT. Further research and animal studies are still required to establish a more detailed and direct mechanistic model.

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Declarations

Competing Interests The authors declare no competing interests.

Ethical Approval and Consent to Participate Not applicable. This manuscript is a narrative review and does not report original research involving human participants, human data, human tissue, or animals.

Consent for Publication Not applicable.

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